Bench Contribution Analysis: Volume vs. Efficiency in NBA Playoffs

Course: DSA 210: Introduction to Data Science

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1. Executive Summary

This project investigates the impact of bench players (non-starters) on NBA Playoff success from 2010 to 2025. The central question is whether "Depth" (scoring volume) or "Efficiency" (quality of contribution) is the primary driver of winning a playoff series.

By scraping and analyzing over 225 playoff series, this study reveals a counter-intuitive finding: **Bench scoring volume is negatively correlated with winning (P<10−39)**. Winning teams typically receive *fewer* points from their bench but significantly higher **Value Over Replacement Player (VORP)** and **Efficiency (BPM)**.

Using these insights, a Machine Learning model (Logistic Regression & Random Forest) was developed to predict playoff series outcomes. The model achieved **90.67% accuracy** on the test set (2020–2025), correctly forecasting the winner in 68 out of 75 recent series.

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2. Motivation & Problem Statement

Basketball analysis often centers on "Star Power" versus "Depth."

- **The "Depth" Argument:** A deep bench protects a team against fatigue and injuries, providing a safety net during the regular season.
- **The "Star" Argument:** In the playoffs, rotations shorten. Stars play 40+ minutes, rendering the bench less significant in terms of volume.

Objective: To move beyond basic counting stats (Points Per Game) and use advanced metrics to mathematically determine the ideal bench composition for a championship contender.

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3. Data & Methodology

3.1 Data Collection

Data was scraped programmatically from **Basketball Reference** using Python libraries (`requests`, `BeautifulSoup`, `pandas`).

- **Scope:** 15 NBA Seasons (2010–2025).
- **Datasets:**
    - Regular Season Player Stats (~7,000 rows).
    - Playoff Advanced Metrics (BPM, VORP).
    - Playoff Series Outcomes (Winner/Loser).

3.2 Terminology & Preprocessing

- **Bench Definition:** A player is classified as "Bench" if they started fewer than 50% of the games they played in that specific season.
- **Differential Calculation:** For every playoff series, we calculated the difference between the two teams:

  $$\text{Diff} = \text{Winner Metric} - \text{Loser Metric}$$

- **Standardization:** Features were scaled using Z-score standardization ($\mu=0, \sigma=1$) to ensure fair comparison between high-magnitude stats (Points) and low-magnitude stats (VORP).

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4. Hypothesis Testing Results

I tested four hypotheses to pinpoint the relationship between bench production and winning.

Hypothesis	Metric Tested	P-Value	Result	Interpretation
H1: Regular Season Depth	Total Bench Points	0.784	**Fail to Reject Null**	Regular season depth has **zero** correlation with playoff success.
H2: Playoff Volume	Bench PPG Diff	3.11×10^{-39}	**Reject Null (Negative)**	Winners get **-2.33 fewer PPG** from their bench than losers.
H3: Playoff Efficiency	Bench BPM Diff	4.48×10^{-4}	**Reject Null (Positive)**	Winning benches are **more efficient** (+0.52 BPM).
H4: Playoff Value	Bench VORP Diff	2.84×10^{-12}	**Reject Null (Positive)**	Winning benches provide **2x the Value** (VORP) of losing benches.

Key Finding: The "Volume Trap." Teams relying on high bench scoring often lose, likely because their starters are underperforming or injured.

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5. Visual Analysis (EDA)

Three advanced visualizations were generated to confirm these statistical findings.

5.1 The "Volume Trap" (Scatterplot)

By plotting **Bench Scoring (X)** vs. **Bench Value (Y)**, I identified a distinct cluster of "Losers" (Red) in the bottom-right quadrant: teams that scored high points but provided low value. *(See `figures/scatterplot_volume_vs_value.png` in repository)*

5.2 Value Distribution (Boxplot)

Winners consistently show a higher median VORP. While losing benches have a wide variance (often dipping into negative value), winning benches are consistent and positive. *(See `figures/boxplot_vorp.png` in repository)*

5.3 Feature Correlation (Heatmap)

A correlation matrix confirmed that `Diff_PPG` is strongly negatively correlated with the target variable `Win`, while `Diff_VORP`is strongly positively correlated. *(See `figures/heatmap_correlation.png` in repository)*

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6. Machine Learning & Prediction

To test the predictive power of these insights, I built a forecasting model.

6.1 Setup

- **Training Set:** 2010–2019 (150 Series).
- **Testing Set:** 2020–2025 (75 Series).
- **Features:** `Diff_PPG`, `Diff_VORP`, `Diff_BPM`.
- **Target:** Binary Classification (1 = Team A Wins, 0 = Team B Wins).

6.2 Model Performance

Model	Accuracy	Key Insight
Logistic Regression	**90.67%**	Coefficient for `Diff_PPG` was **-4.00**, confirming volume is a massive penalty.
Random Forest	**89.33%**	ROC Analysis showed strong separation between classes. Balanced Confusion Matrix.

6.3 Real-World Validation

The model correctly predicted the winner in **14 of the last 15** major series (Conference Finals & NBA Finals), often with high confidence (>75%).

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7. Limitations & Future Work

Limitations:

1. **Metric Limitations:** Box-score derived metrics (BPM/VORP) may miss intangible defensive contributions.
2. **Binary "Bench" Definition:** The <50% starts rule does not account for "Sixth Men" who play starter minutes.
3. **Context:** The model does not know about mid-series injuries or coaching adjustments.

Future Work:

1. **Lineup Analysis:** Scraping play-by-play data to measure 5-man unit Net Ratings instead of individual averages.
2. **Live Dashboard:** Implementing a Streamlit app to update win probabilities after every game in a series.

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8. AI Usage Disclosure

In compliance with academic integrity guidelines, the following section details the use of Artificial Intelligence tools in this project.

Tools Used:

- **Large Language Models (LLMs):** Specifically used for code debugging, syntax reference, and technical explanation drafting.

Specific Use Cases:

1. **Code Assistance:** AI was used to generate templates for standard Data Science libraries (`pandas` merges, `matplotlib`plotting functions, and `scikit-learn` pipeline syntax).
2. **Debugging:** AI assisted in resolving a specific `KeyError` during the visualization phase by identifying that raw columns were missing from the feature engineering step.
3. **Concept Clarification:** AI provided clear, mathematical explanations for "Standardization" and "Logistic Regression Sigmoid Functions," which helped refine the methodology section of this report.
4. **Formatting:** AI helped structure the Markdown files (README and Final Report) to ensure clean, professional formatting.

Statement of Originality: While AI was used as a coding assistant and technical reference, **the core hypothesis generation, data collection strategy, interpretation of**

statistical results, and final conclusions are entirely my own work. The decision to focus on "Efficiency vs. Volume" and the selection of VORP as the critical metric were driven by my personal domain knowledge of basketball.